MODELING FULL-EMBRYO MORPHOGENESIS USING LOCAL MESOSCOPIC PROPERTIES

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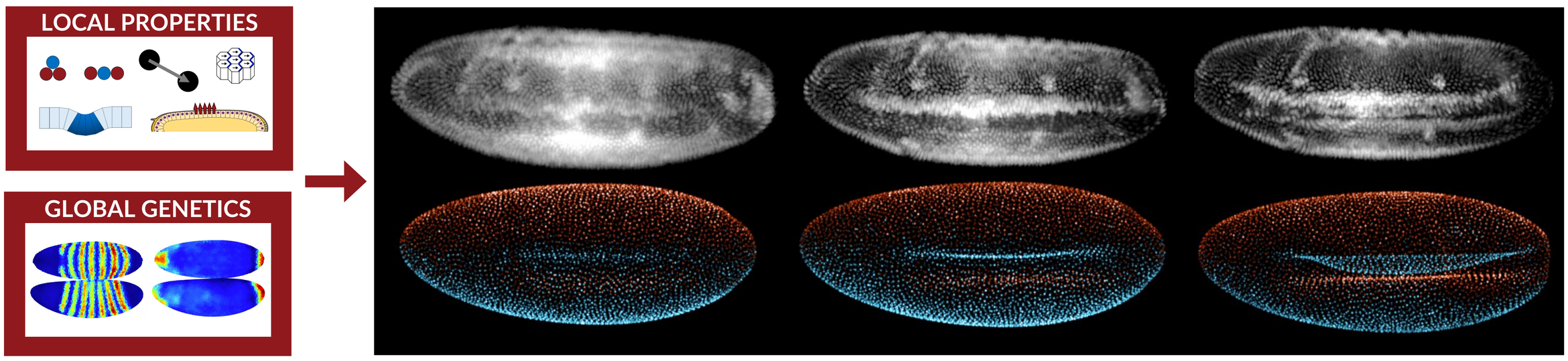
GRAPHICAL ABSTRACT



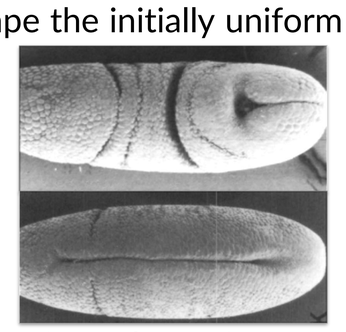
MOTIVATION

To make multicellular life feasible, cells must be able to consistently produce precise, large-scale morphological structures.

Most scientific models isolate single events, single biological mechanisms, or single spatial domains, limiting insight into cellular interaction across time, space and length scales.

INTRODUCTION

Morphogenesis is the study in the formation of physical shape. Drosophila gastrulation is a widely studied morphogenetic event consisting of intricately coordinated movement of 5000 cells. These actions reshape the initially uniform embryonic cell sheet into a characteristic shape with distinct domains.



Mesoscopic properties lies between subcellular and whole-embryo properties. These features include protein-aggregates, cell adhesion and local anisotropic reshaping, among other things. Importantly, all these different properties are dynamically coupled through numerous separate biophysical mechanisms.

Modeling this mesoscale, we implement an agent-based approach; explicitly defining the polarities, adhesion and deformations. We simulate cells across the entire embryo simultaneously cells reacting to- and altering their local environment.

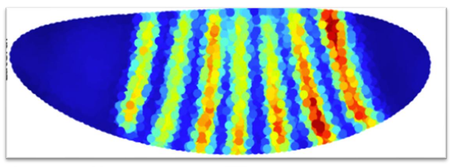
METHODS

We model cell centers as point particles with explicit apical-basal and planar polarity vectors. Each cell has a short-range repulsion and a long-range attraction to their neighbors, defining the cell-cell potential:

$$\boldsymbol{u}_{ij} = \mathrm{e}^{r_{ij}} - \boldsymbol{S}_{ij}\mathrm{e}^{r_{ij}}$$

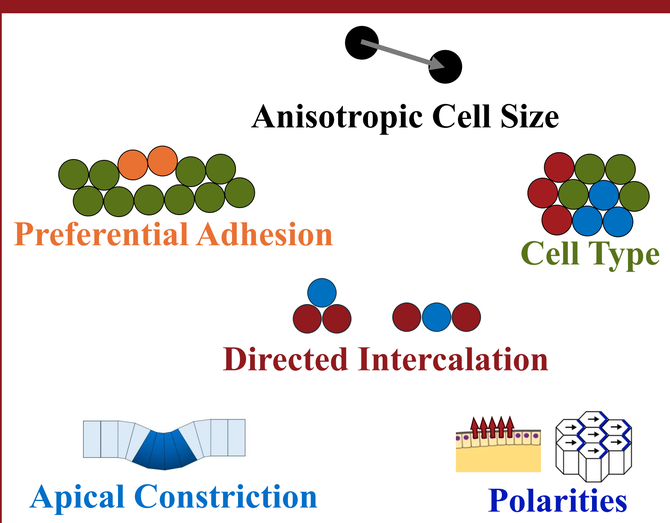
Where $\boldsymbol{S}_{ij}$ is describing the dynamic attraction between the cells by the local properties.

Parameters, initial- and boundary-conditions are found by virtue of transcriptomics data, defining areas of similar dynamics.



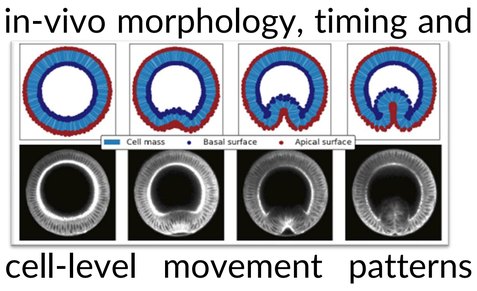
The potential is summed for each cell ($\boldsymbol{U}_i = \sum_{\{j\}} u_{ij}$) every step. Then the forces are found ($\boldsymbol{F} = -\nabla \boldsymbol{U}$), and the whole system updated by overdamped Langevin dynamics ($\Delta \boldsymbol{X} = \boldsymbol{F}\mathrm{d}t + \boldsymbol{\eta}\mathrm{d}t$).

DETOUR: LOCAL PROPERTIES

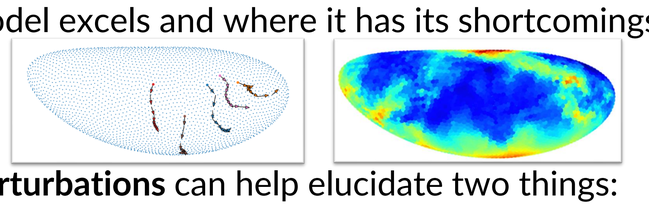


PRELIMINARY RESULTS

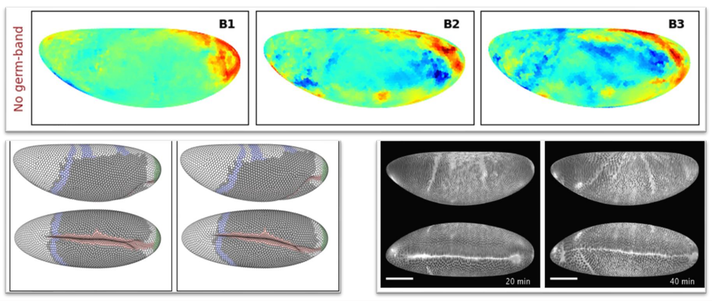
Qualitatively, the model recapitulates major gastrulation events. Simulated embryos closely match observed in-vivo morphology, timing and dynamics. An example is the closure of the Ventral Furrow



Quantitatively, cell-level movement patterns are aligning well in most parts. The cumulative per-cell-error pinpoints the domains where our model excels and where it has its shortcomings



Perturbations can help elucidate two things: 1) Perturbing one domain can probe how other spatial domains reacts/interacts. 2) Reproducing mutant phenotypes helps validate the biological validity and generalizability of the model.



CONCLUSION & QUESTIONS

- Complex timing and spatial order emerges without explicit temporal regulation or long-range communication.
- Our mesoscale approach enable perturbations that validate the model, probe inter-domain interactions, and yield testable predictions.

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